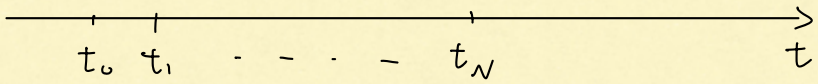
$$|\psi(t)\rangle = U(t-t_0)|\psi(t_0)\rangle \quad , \quad U(t-t_0) = e^{-iH(t-t_0)/\hbar}$$

↳ 左乘 $\langle \vec{r}|$: $\psi(\vec{r},t) = \int \langle \vec{r}|U(t-t_0)|\vec{r}_0\rangle\langle \vec{r}_0|\psi(t_0)\rangle \, d\vec{r}_0$

$$= \int K(\vec{r},t;\vec{r}_0,t_0)\,\psi(\vec{r}_0,t_0)\,d\vec{r}_0$$

↳ propagator



不一定是均分

$$\max\{t_n - t_{n-1}\} \longrightarrow 0$$

$$K(\vec{r},t;\vec{r}_0,t_0) = \langle \vec{r}|U(t,t_0)|\vec{r}_0\rangle = \langle \vec{r}|U(t,t_1)U(t_1,t_0)|\vec{r}_0\rangle$$

$$= \int \langle \vec{r}|U(t,t_1)|\vec{r}_1\rangle\langle \vec{r}_1|U(t_1-t_0)|\vec{r}_0\rangle \, d\vec{r}_1$$

$$= \int \langle \vec{r}|U(t,t_2)|\vec{r}_2\rangle\langle \vec{r}_2|U(t_2-t_1)|\vec{r}_1\rangle\langle \vec{r}_1|U(t_1-t_0)|\vec{r}_0\rangle \, d\vec{r}_1\, d\vec{r}_2$$

$$= \int K(\vec{r},t;\vec{r}_2,t_2)\,K(\vec{r}_2,t_2;\vec{r}_1,t_1)\,K(\vec{r}_1,t_1;\vec{r}_0,t_0)\,d\vec{r}_1\, d\vec{r}_2$$

$$\rightsquigarrow K(\vec{r},t;\vec{r}_0,t_0) = \int \left[\prod_{n=1}^{N} K(\vec{r}_n,t_n;\vec{r}_{n-1},t_{n-1})\right] d\vec{r}_1 \cdots d\vec{r}_{N-1}$$

$$= \int \left[\prod_{n=1}^{N} \mathcal{K}(\vec{r}_n,t_n;\vec{r}_{n-1},t_{n-1})\right] d\vec{r}_1 \cdots d\vec{r}_{N-1}$$

↳ 只是换个字体

记 $\Delta t_n = t_n - t_{n-1}$

$$\mathcal{K}(\vec{r}_n,t_n;\vec{r}_{n-1},t_{n-1}) = \langle \vec{r}_n|U(t_n,t_{n-1})|\vec{r}_{n-1}\rangle = \langle \vec{r}_n|e^{-iH\Delta t_n/\hbar}|\vec{r}_{n-1}\rangle$$

$$e^{-iH\Delta t_n/\hbar} = e^{-i\left(\frac{p^2}{2m}+V(\vec{R},t_n)\right)\Delta t_n/\hbar} \overset{\Delta t_n\to 0}{=\!=} e^{-\frac{i}{\hbar}V(\vec{R},t_n)\Delta t_n}\, e^{-\frac{i}{\hbar}\frac{p^2}{2m}\Delta t_n}$$

↳ 从这一步起，$K \neq \mathcal{K}$，所以换记号

$$\mathcal{K}(\vec{r}_n,t_n;\vec{r}_{n-1},t_{n-1}) = \langle \vec{r}_n|e^{-\frac{i}{\hbar}V(\vec{R},t_n)\Delta t_n}\, e^{-\frac{i}{\hbar}\frac{p^2}{2m}\Delta t_n}|\vec{r}_{n-1}\rangle$$

$$= e^{-\frac{i}{\hbar}V(\vec{r}_n,t_n)\Delta t_n}\,\langle \vec{r}_n|e^{-\frac{i}{\hbar}\frac{p^2}{2m}\Delta t_n}|\vec{r}_{n-1}\rangle$$

$$= \int e^{-\frac{i}{\hbar}V(\vec{r}_n,t_n)\Delta t_n}\,\langle \vec{r}_n|\vec{p}_n\rangle\langle \vec{p}_n|e^{-\frac{i}{\hbar}\frac{p^2}{2m}\Delta t_n}|\vec{r}_{n-1}\rangle \, d\vec{p}_n$$

$$= \int e^{-\frac{i}{\hbar}V(\vec{r}_n,t_n)\Delta t_n}\,\langle \vec{r}_n|\vec{p}_n\rangle\, e^{-\frac{i}{\hbar}\frac{p_n^2}{2m}\Delta t_n}\,\langle \vec{p}_n|\vec{r}_{n-1}\rangle \, d\vec{p}_n$$

$$= \frac{1}{(2\pi\hbar)^3}\int e^{-\frac{i}{\hbar}V(\vec{r}_n,t_n)\Delta t_n}\, e^{\frac{i}{\hbar}\vec{p}_n\cdot\Delta\vec{r}_n}\, e^{-\frac{i}{\hbar}\frac{p_n^2}{2m}\Delta t_n}\, d\vec{p}_n$$

$$\Delta\vec{r}_n = \vec{r}_n - \vec{r}_{n-1}$$

$$\dot{\vec{r}}_n = \Delta\vec{r}_n/\Delta t_n$$

$$= \frac{1}{(2\pi\hbar)^3} \int e^{\frac{i}{\hbar}\left[\vec{p}_n \cdot \frac{\Delta \vec{r}_n}{\Delta t_n} - H(\vec{r}_n, \vec{p}_n, t_n)\right]\Delta t_n} \, d\vec{p}_n$$

$$= \frac{1}{(2\pi\hbar)^3} \int e^{\frac{i}{\hbar}\left[\vec{p}_n \cdot \dot{\vec{r}}_n - H(\vec{r}_n, \vec{p}_n, t_n)\right]\Delta t_n} \, d\vec{p}_n$$

$$= \frac{1}{(2\pi\hbar)^3} \int e^{\frac{i}{\hbar} L(\vec{r}_n, \vec{p}_n, t_n)\Delta t_n} \, d\vec{p}_n$$

$$K(\vec{r}, t; \vec{r}_0, t_0) = \int \left[\prod_{n=1}^{N} K(\vec{r}_n, t_n; \vec{r}_{n-1}, t_{n-1})\right] d\vec{r}_1 \cdots d\vec{r}_{N-1}$$

$$= \int \prod_{n=1}^{N} \left[\frac{1}{(2\pi\hbar)^3} \int e^{\frac{i}{\hbar} L(\vec{r}_n, \vec{p}_n, t_n)\Delta t_n} d\vec{p}_n\right] d\vec{r}_1 \cdots d\vec{r}_{N-1}$$

$$= \frac{1}{(2\pi\hbar)^{3N}} \int \left[\int e^{\frac{i}{\hbar}\sum_{n=1}^{N} L(\vec{r}_n, \vec{p}_n, t_n)\Delta t_n} d\vec{p}_1 \cdots d\vec{p}_N\right] d\vec{r}_1 \cdots d\vec{r}_{N-1}$$

$$= \frac{1}{(2\pi\hbar)^{3N}} \int e^{\frac{i}{\hbar}\int L(\vec{r}, \vec{p}, \tau) d\tau} \, d\vec{r}_1 \cdots d\vec{r}_{N-1} \, d\vec{p}_1 \cdots d\vec{p}_N$$

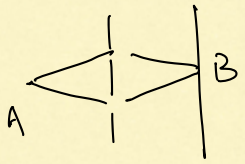
$$= \frac{1}{(2\pi\hbar)^{3N}} \int D\vec{r} \, D\vec{p} \, e^{i\frac{S(\vec{r}, \vec{p}, t)}{\hbar}}$$

路径积分 Path-Integral

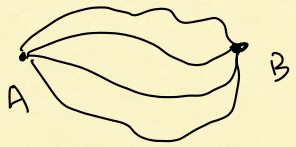
波动
矩阵 } 基于算符的正则方法
路径积分

等价，但不同问题的方便程度不同

① 双缝干涉



② ∞ 板，∞ 孔



振幅 $\sim \int [\text{所有路径}]\, e^{iS}$

正则方法：显式的幺正性
数学上易严格化
第一原理明确，可处理疑难杂症

路径积分：显式的 Lorentz 不变性
优雅地处理规范不变性，非微扰问题

1.1 路径积分：出发点，计算目标

* 出发点：正则量子化的系统：等时量子化

$[Q_a, Q_b]=0 \implies Q_a|q\rangle = q_a|q\rangle$

$[P_a, P_b]=0 \implies P_a|p\rangle = P_a|p\rangle$

$[Q_a, P_b] = i\delta_{ab} \implies \langle q|p\rangle = \prod_a \frac{1}{\sqrt{2\pi}} e^{iq_a p_a}$

$\hookrightarrow = i\partial/\partial Q_b$

$\langle q'|q\rangle = \prod_a \delta(q'_a - q_a) \equiv \delta(q'-q)$

$\int \prod_a dq_a |q\rangle\langle q| = 1$

$\langle p'|p\rangle = \prod_a \delta(p'_a - p_a) \equiv \delta(p'-p)$

$\int \prod_a dp_a |p\rangle\langle p| = 1$

* 随时间演化：（P_a，$|p;t\rangle$ 相似）

$$Q_a(t) \equiv e^{iHt} Q_a e^{-iHt}, \quad |q;t\rangle = e^{iHt}|q\rangle$$

$$Q_a(t)|q;t\rangle = e^{iHt} Q_a|q\rangle = q_a|q;t\rangle$$

$$H \equiv H(Q,P) = e^{iHt} H(Q,P) e^{-iHt} = H(Q(t), P(t))$$

H中算符顺序：所有 Q_a 在 P_a 左边

* 要算：跃迁振幅 $\langle q';t'|q;t\rangle$

i.e. 在 t 时刻制备系统处于态 $|q;t\rangle$，让系统按 H 演化到 t' 时刻，再测量得态 $|q';t'\rangle$ 的概率幅度。 内积怎么算？ $\implies$ 用 e^{iHt} 变到同一时刻

$$\langle q';t'|q,t\rangle = \langle q';t|e^{-iH(t-t')}|q;t\rangle$$

困难：$H=H(Q(t),P(t))$ 在指数上

若计算 $\langle q';t|H(t-t')|q,t\rangle$ 很简单：

$$\parallel$$

$$\int\prod_a dp_a \langle q't|H(Q,P)(t-t')|pt\rangle\langle pt|qt\rangle$$

$Q\to q$ $\quad$ $P\to p$

但 e^{-iHt} 展开后，不是所有 Q在左 P在右，所以不行

* 解决方案：

把时间分成 ∞ 个 $\frac{1}{\infty}$ 片段：$\quad d\tau = \tau_{k+1}-\tau_k = \frac{t'-t}{N+1}$

每个片段 $e^{-iH d\tau} \simeq 1 - iH d\tau + O(d\tau^2)$

满足 Q在左 P在右 $\Longrightarrow$ 可插入 $\int\prod_a dp_a |p\tau_k\rangle\langle p\tau_k|$

即：$\langle q't'|qt\rangle = \int\left(\prod_a dq_{1a}\right)\left(\prod_a dq_{2a}\right)\cdots\left(\prod_a dq_{Na}\right)$

$q_a(\tau_k)\equiv q_{ka}$

$$\times\ \langle q't'|q_N\tau_N\rangle\langle q_N\tau_N|q_{N-1}\tau_{N-1}\rangle\cdots\langle q_1t_1|qt\rangle$$

1.2 推导路径积分公式

* 计算其中一项：$\langle q_{k+1}\tau_{k+1}|q_k\tau_k\rangle = \int\prod_a dp_a\langle q_{k+1};\tau_k+d\tau|p_k\tau_k\rangle\langle p_k\tau_k|q_k\tau_k\rangle$

$$= \int\prod_a dp_a\langle q_{k+1};\tau_k|e^{-iH(\overset{\approx q_k}{q_{k+1}},q_k)d\tau}|p_k\tau_k\rangle\langle p_k\tau_k|q_k\tau_k\rangle \qquad \langle p_k\tau_k|q_k\tau_k\rangle = \prod_a\frac{1}{\sqrt{2\pi}}e^{-q_{ak}p_{ak}}$$

$$= \int\prod_a\frac{dp_a}{2\pi}\exp\left\{-iH(q_k,p_k)d\tau + i\sum_a\underbrace{(q_{k+1,a}-q_{ka})}_{\text{求得像}\ \dot{q}\,d\tau}p_{ak}\right\}$$

* 所有项乘积：路径积分公式

$$\langle q't'|qt\rangle = \int_{\substack{q_a(t)=q_a\\ q_a(t')=q_a'}}\prod_{\tau,a}dq_a(\tau)\prod_{\tau,b}\frac{dp_b(\tau)}{2\pi}\,e^{iI[q,p]}$$

其中 $I[q,p]=\int_t^{t'}d\tau\left\{\sum_a\dot{q}_a(\tau)p_a(\tau)-H(q(\tau),p(\tau))\right\}$

形式上是作用量，但仍是 q,p 的泛函

* 编时乘积的路径积分

设 P在左，Q在右

$$\langle q't' | T\{ O_A(P(t_A), Q(t_A))\, O_B(P(t_B), Q(t_B)) \cdots \} | q t \rangle$$

$$= \int_{\substack{q_a(t)=q_a \\ q_a(t')=q_a'}} \prod_{\tau, a} dq_a(\tau) \prod_{\tau, b} \frac{dp_b(\tau)}{2\pi} e^{iI[q,p]}\, O_A(p(t_A), q(t_A))\, O_B(p(t_B), q(t_B)) \cdots$$